AFFINE FINITE TYPE SPACES IMMERSE INTO AFFINE SPACE OVER LOCALLY SEPARATED QS TARGETS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3276-3291, following the statement at 3268-3274.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Locally separated diagonal is an immersion. Morphisms of Spaces, Lemma spaces-morphisms-lemma-semi-diagonal (cross-chapter, cited at ~3281-3283). U -> X x U is an immersion in the limit.
- Eventual closedness for finite type. lemma-finite-type-eventually-closed (same chapter, cited at ~3284-3286). U -> X x U_i is an immersion for some i.

Source reference. Original lemma 0828: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1452–1452. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KO: focused-seeds/geometry/S9-spaces-limits-embedding-into-affine-over-ls-qs/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : U \to X$ be a morphism of algebraic spaces over S. Assume U is an affine scheme, f is locally of finite type, and X quasi-separated and locally separated. Then there exists an immersion $U \to \mathbf{A}^n_X$ over X.*

Proof. Say $U = \mathrm{Spec}(A)$. Write $A = \mathrm{colim}\, A_i$ as a filtered colimit of finite type $\mathbf{Z}$-subalgebras. For each i the morphism $U \to U_i = \mathrm{Spec}(A_i)$ induces a morphism

$$U \longrightarrow X \times U_i$$

over X. In the limit the morphism $U \to X \times U$ is an immersion as X is locally separated, see Morphisms of Spaces, Lemma 03KO. By Lemma 0828 we see that $U \to X \times U_i$ is an immersion for some i. Since U_i is isomorphic to a closed subscheme of $\mathbf{A}^n_{\mathbf{Z}}$ the lemma follows. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.